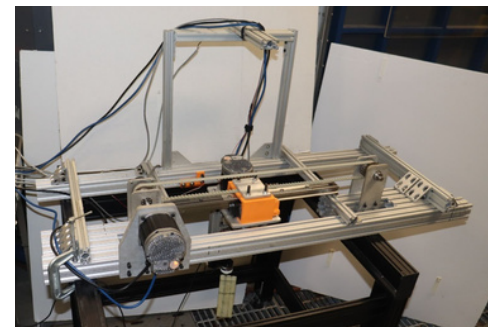
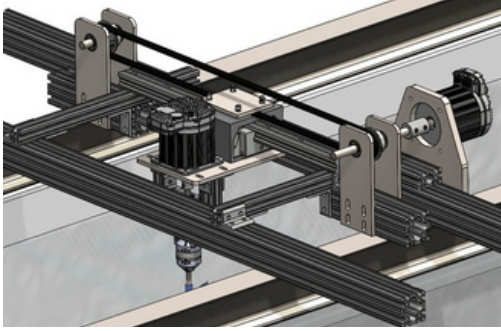


PITCH AND HEAVE SETUP FOR BIO-PROPULSION STUDIES



What?

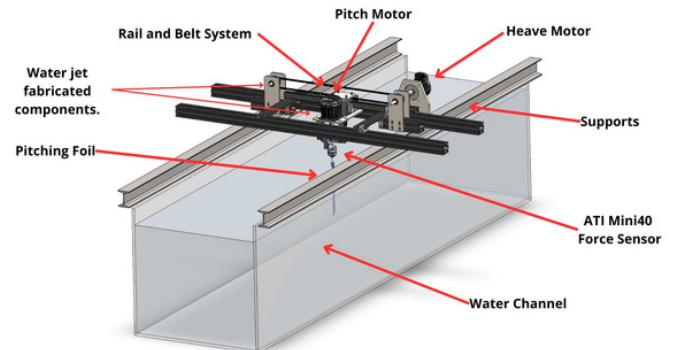
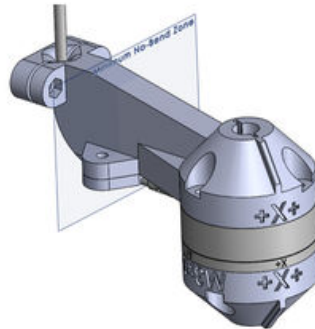
- Designed and built a robotic swimmer system for a water channel to mimic fish fin swimming and study how energy from ocean waves can power and propel marine vessels.

How?

- Designed and built the setup using SOLIDWORKS, CNC, waterjet cutting, and 3D printing.
- Applied GD&T to all drawings.
- Integrated systems such as PIV system, 6-axis force sensor, NI DAQ data acquisition card, high-torque motors.

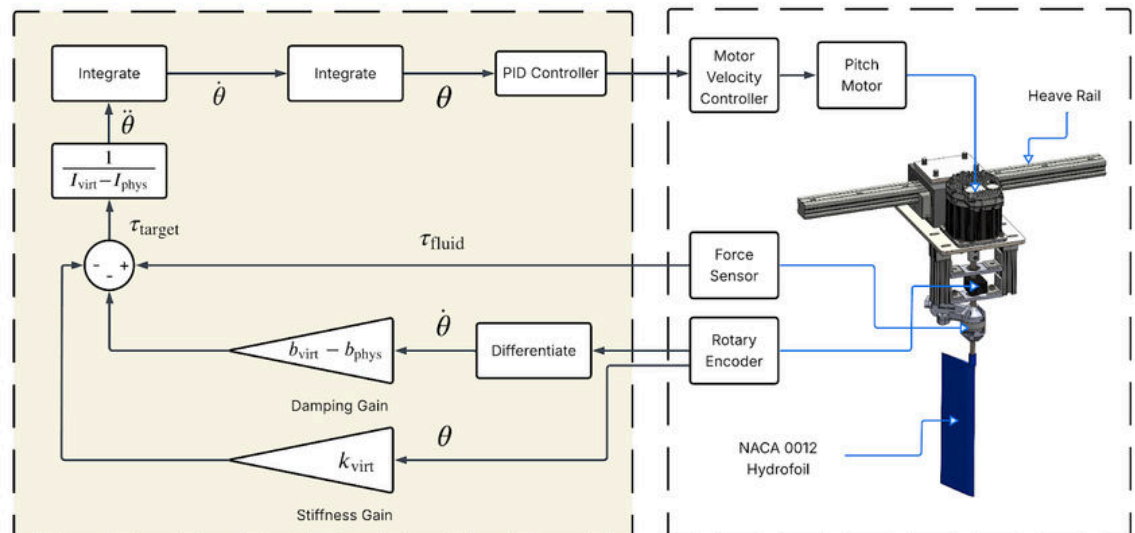
Results

- Gathered and analyzed 35 data sets using MATLAB and NI DAQ for tests.
- Traced motion error to backlash and redesigned the assembly, reducing error by 5x over previous designs.
- Reduced operational noise by 80% through optimized control systems.

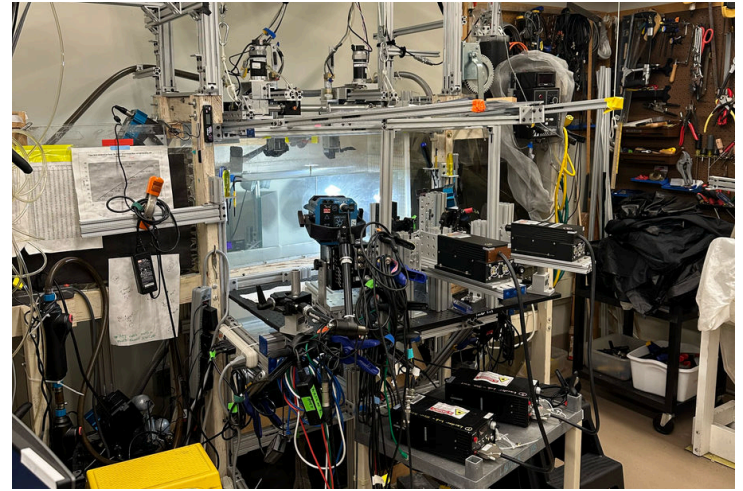


Cyber System

Physical System



FISH ROBOT SWIMMING EFFICIENCY - HARVARD UNIVERSITY



What?

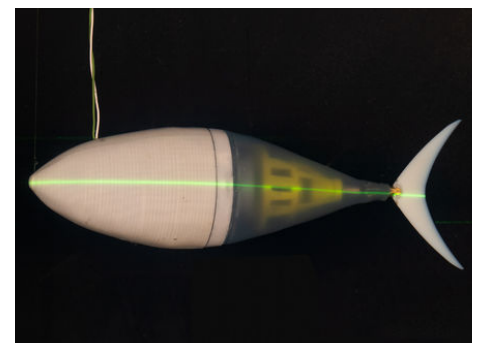
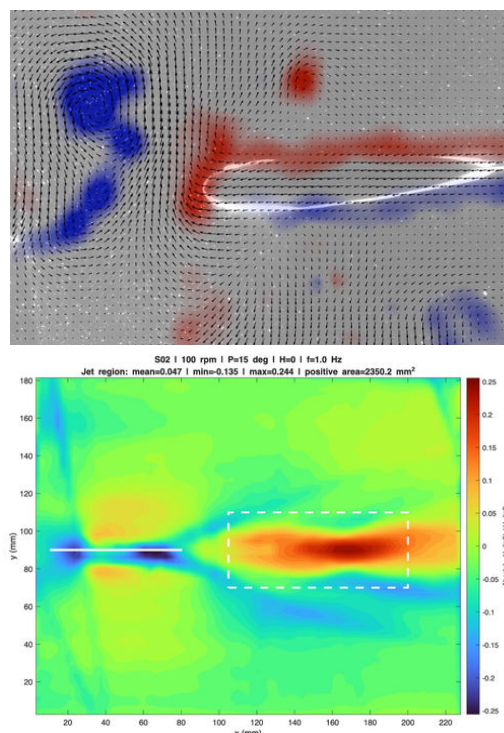
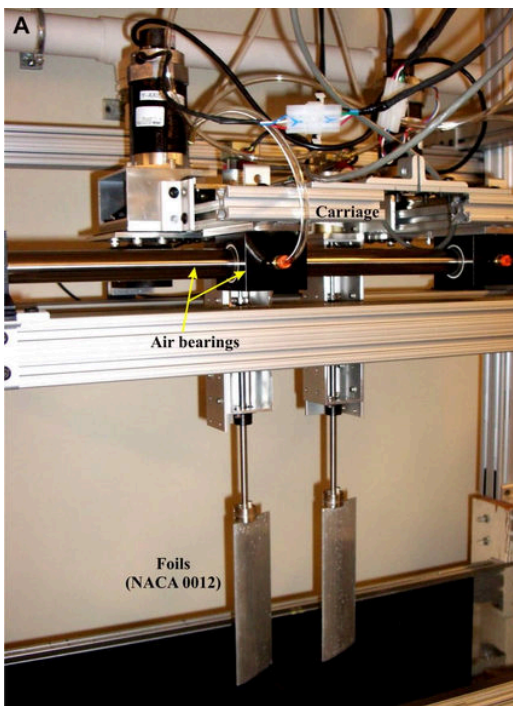
- Studied how a fish swimming in line behind a leading swimmer gains thrust from its wake without actively swimming, with applications to ships and underwater drones.
- Measured how frequency, pitch amplitude, and flow speed affect swimming and station-holding performance.

How?

- Operated and redesigned a robotic flapper with air bearings, LabVIEW, PID control, and NI-DAQ.
- Ran hydrofoil and passive-fish experiments across flow conditions.
- Performed particle image velocimetry (PIV); processed wake data in MATLAB and PIVlab.
- Built 2-D tracking and 3-D reconstruction pipelines in MATLAB and R.

Results

- Presented findings at the ONR MURI Annual Review Meeting (Sep 2026) and the APS Division of Fluid Dynamics Meeting (Nov 2026).
- Quantified swimming kinematics across flow conditions, which can inform ship propulsion and underwater drone design.



CARBON FIBER SUSPENSION SYSTEM FOR FORMULA SAE



What?

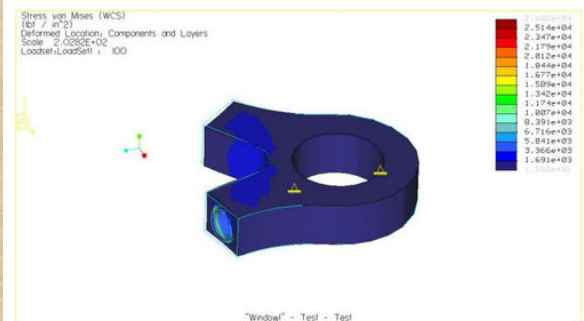
- Designed and manufactured a carbon fiber-aluminum hybrid suspension control arm for a Formula SAE vehicle.
- Replaced steel members to reduce mass while maintaining required load capacity.

How?

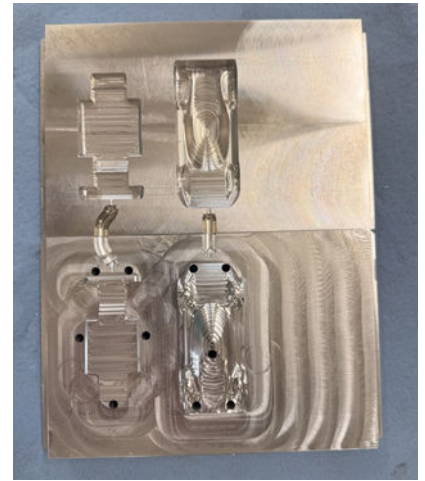
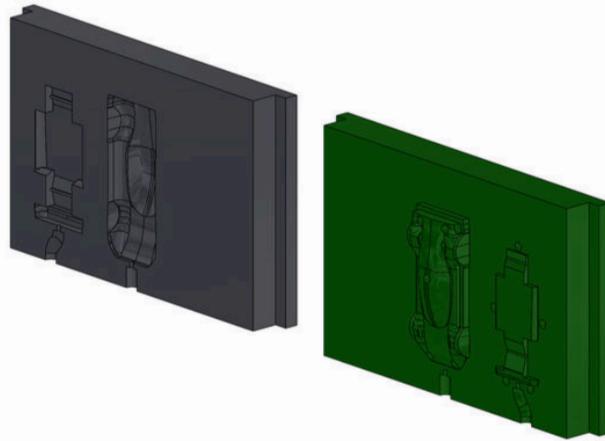
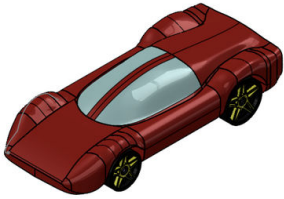
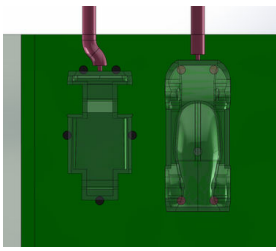
- Modeled geometry in SolidWorks and conducted FEA to evaluate stress and deflection under braking and cornering loads.
- Machined aluminum end joints and bonded them to carbon fiber tubes using structural epoxy.

Results

- Reduced component mass compared to steel baseline.
- Measured strain and load response were consistent with FEA predictions.
- Component sustained expected loads without material failure or joint separation.



RACE CAR INJECTION MOLDING PROJECT



What?

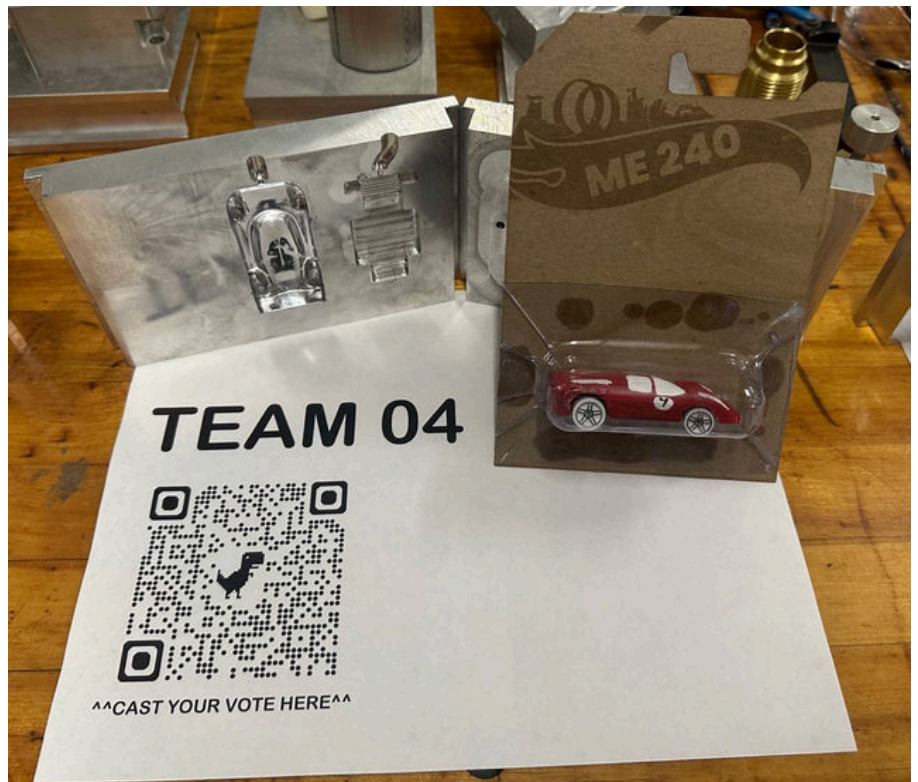
- Designed and manufactured a two-part aluminum injection mold for a race car body component.
- Developed a complete molding workflow including part design, gating, and runner system.

How?

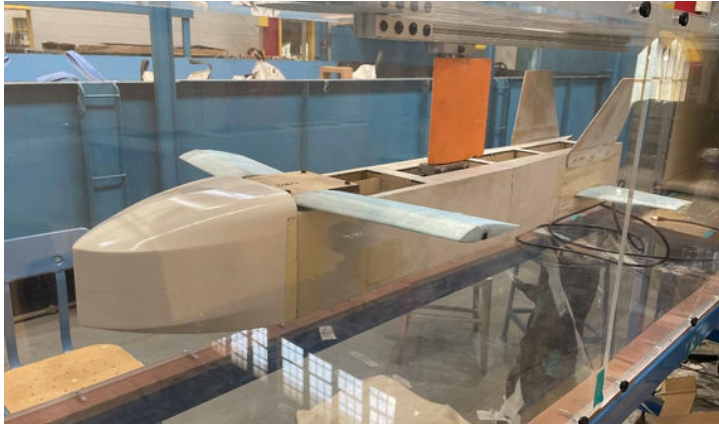
- Modeled part and mold assembly in SolidWorks.
- Fabricated mold using CNC machining and waterjet cutting.
- Applied GD&T to ensure dimensional accuracy and proper part ejection.
- Conducted molding trials and evaluated part quality and repeatability.

Results

- Successfully produced molded race car components with consistent geometry and surface finish.
- Achieved proper mold alignment and clean part release.
- Validated manufacturability through multiple successful production cycles.



WIND TUNNEL DRONE TESTING - AEROTARGETS INTERNATIONAL



What?

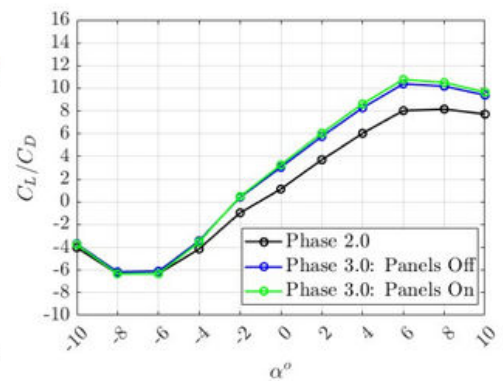
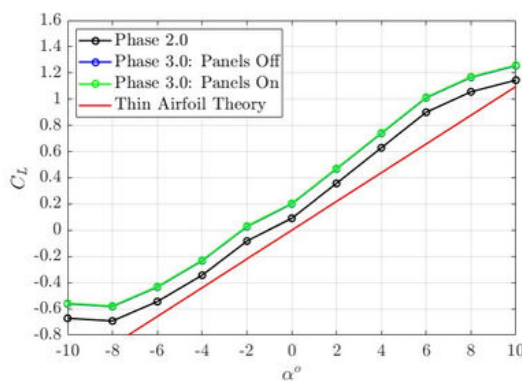
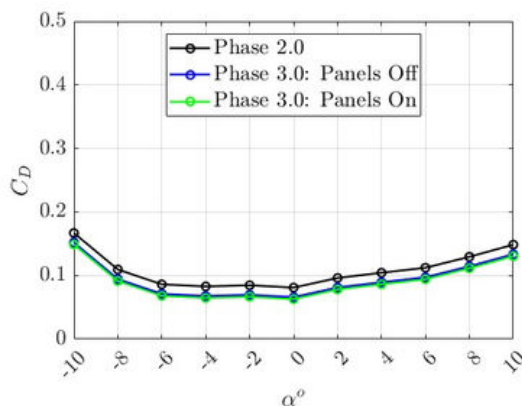
- Tested a scaled-down military drone in a wind tunnel to measure its aerodynamic performance.
- Split the experiment into 3 phases focusing on (1) preliminary aerodynamic stability studies, (2) a scaled up model at a higher Re used for extensive drag tests with panels off and (3) with panels on.

How?

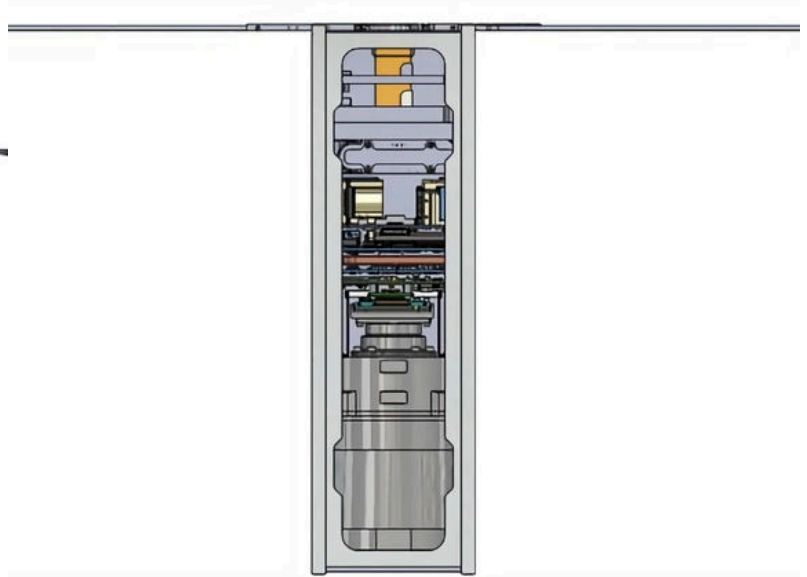
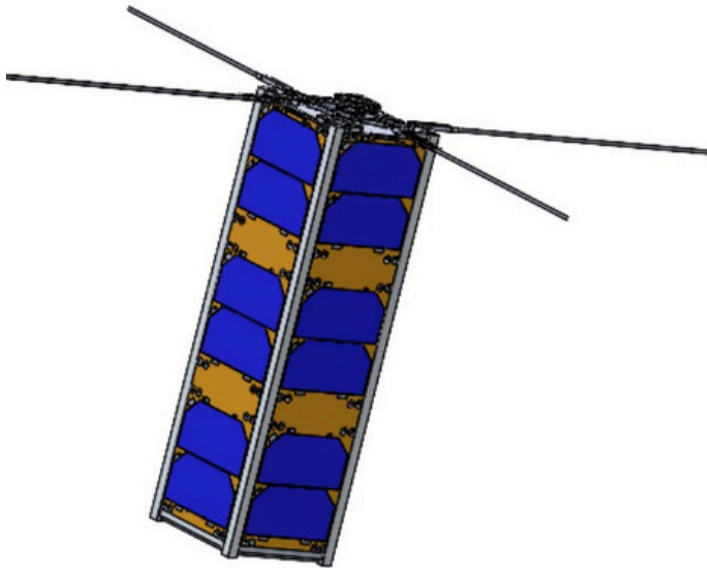
- Built a four-bar linkage structure that allows every test angle to be set without re-rigging.
- Used SolidWorks to scale down and alter model for compatibility with force sensor / wind tunnel.
- Utilized an NI DAQ card and MATLAB to acquire and process data.

Results

- Phase 1 revealed that the model is dynamically stable in both pitch and roll motions around its CG and weakly stable in yaw.
- Phase 2 and 3 showed a lower drag coefficient, increasing its L/D ratio.
- Communicated with AeroTargets engineers to recommend fuselage design modifications, achieving a 10% drag reduction.



OSPREY CUBESAT - LEHIGH UNIVERSITY'S FIRST NANOSATELLITE



What?

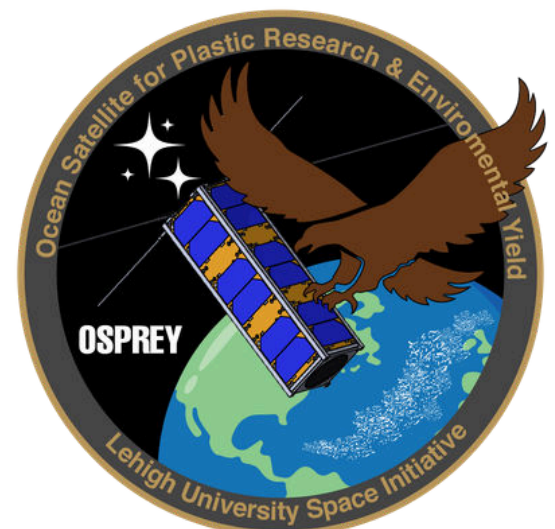
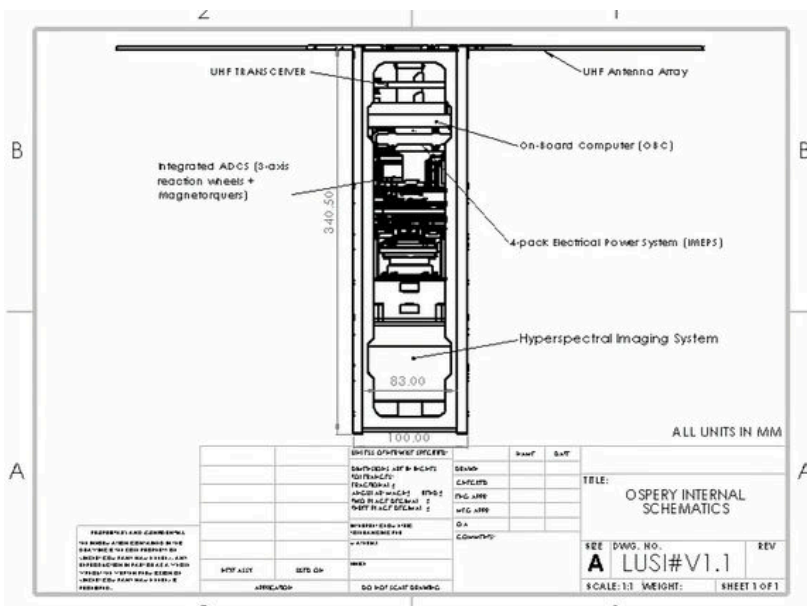
- Led the design of OSPREY, Lehigh University's first CubeSat, a 3U satellite built to spot ocean plastic from orbit.
- Specified a VS/SWIR camera system able to resolve plastic within a 1-square-meter area.

How?

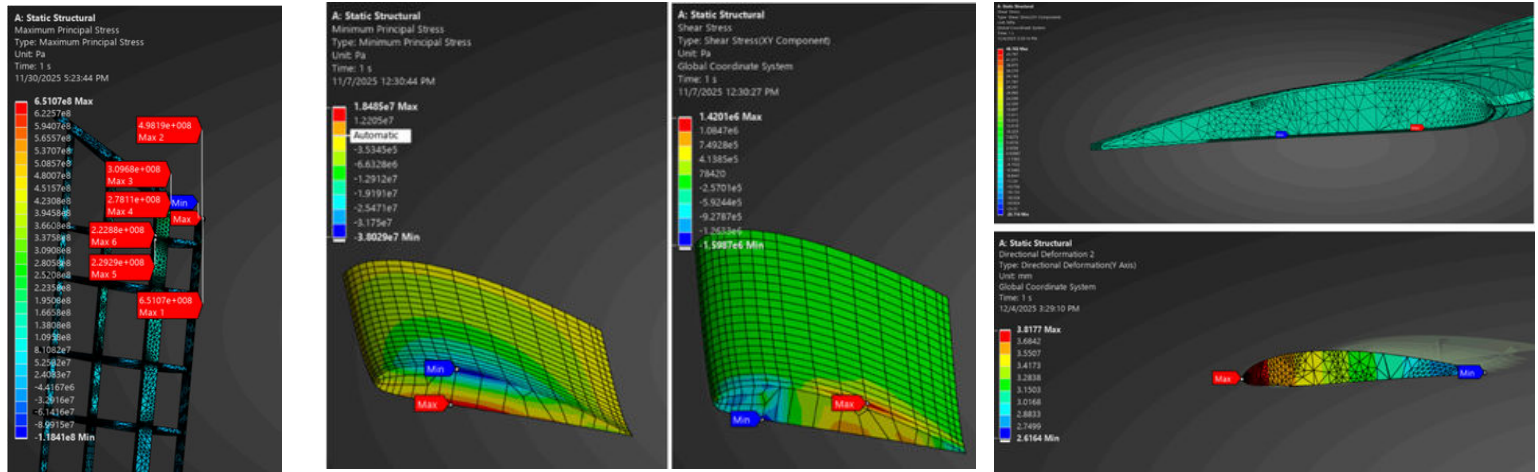
- Led CAD modeling in SolidWorks, optimizing the center of mass and moments of inertia.
- Used GD&T principles, applied material selection, mass budgeting, to ensure tolerance and compliance with NASA standards.
- Managed a team of 6 students and coordinated with other sub-teams to integrate deployable solar cells and antenna systems

Results

- Raised \$150,000 in funding from Lehigh alumni and sponsors.
- Optimized the mass budget by 23% and increased payload volume by 33%.
- Successfully submitted a proposal for NASA's 2024 CubeSat Student Launch Initiative.



AEROELASTIC WING DESIGN - LEHIGH UNIVERSITY



What?

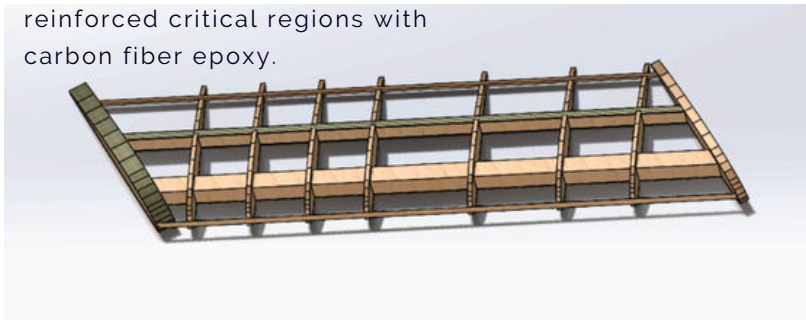
- Designed and tested three balsa wood wings to find how much weight a wing can lose before it becomes unstable in flight.
- Compared straight, swept-forward, and rib-and-strut designs (Gottingen 398 airfoil).
- Evaluated torsional stiffness, aeroelastic coupling, and divergence speed under a 100 N/m distributed lift load.
- Fabricated rib-strut structure using a laser cutter and reinforced critical regions with carbon fiber epoxy.

How?

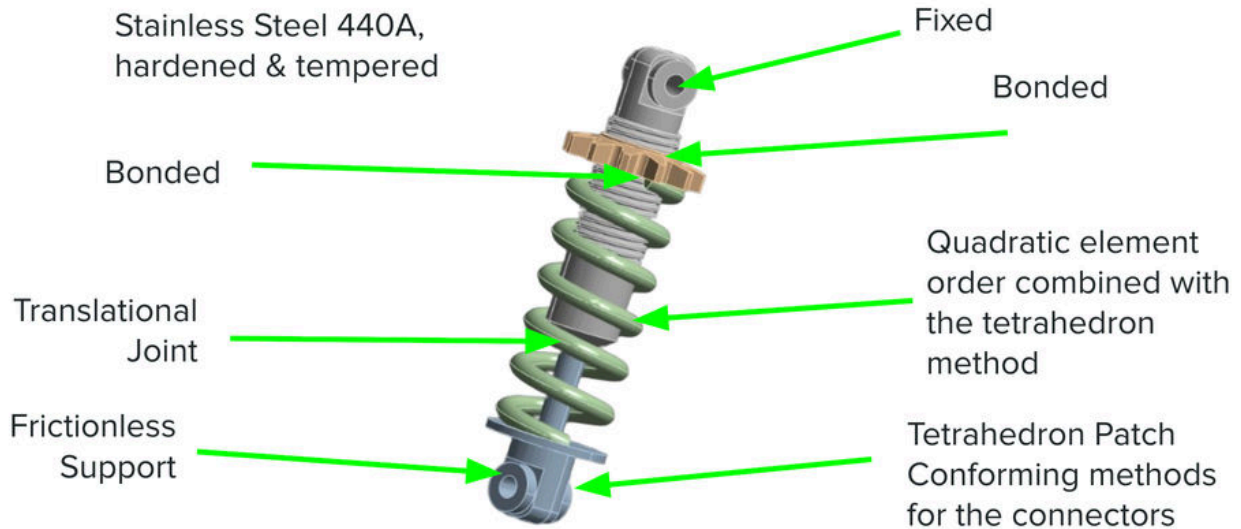
- Modeled geometry in SolidWorks.
- Performed structural analysis using ANSYS Workbench FEA to extract deflections and stresses.
- Computed twist, torsional stiffness, shear center, and divergence speed in MATLAB.
- Conducted wind tunnel testing.

Results

- Showed that aggressive weight reduction (to 0.247 N) led to a ~98% stiffness drop, lowering divergence speed to 26.2 m/s.
- Identified the base shear interface as the critical failure location during wind tunnel testing (~90 mph).
- Established quantitative relationship between mass reduction, stiffness penalty, and aeroelastic stability.



SHOCK ABSORBER - FEA MODAL ANALYSIS



What?

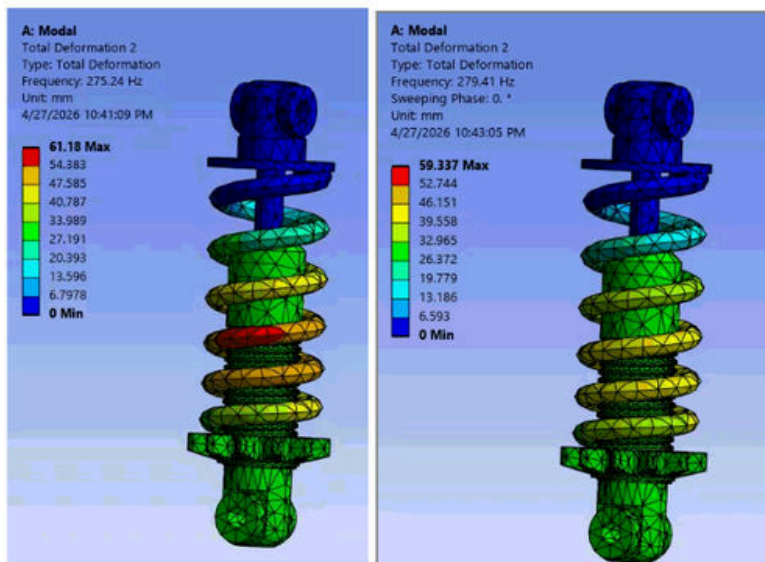
- Performed modal analysis on a scaled shock absorber assembly to study its natural vibration behavior.
- Tested how damping affects vibration decay and stability.

How?

- Set up the model in ANSYS with fixed, bonded, frictionless, and translational joint boundary conditions.
- Used Stainless Steel 440A and quadratic tetrahedral mesh elements.
- Tested damping ratios from 0.05 to 0.3.

Results

- Natural frequencies stayed around 152–712 Hz.
- Higher damping caused faster vibration decay without major frequency shifts.
- Results showed the system is mainly mass–stiffness dominated.
- Fixed contact issues by improving boundary conditions and using a translational joint where sliding motion was needed.



Damping off

Damping on

0.05 damping ratio

Mode	Damped Frequency [Hz]	Stability [Hz]	Modal Damping Ratio	Logarithmic Decrement
1	151.89	-1.551	1.0211e-002	-6.4161e-002
2	355.23	-5.3157	1.4963e-002	-9.4023e-002
3	367.36	-5.5009	1.4994e-002	-9.4222e-002
4	414.7	-6.1958	1.4939e-002	-9.3874e-002
5	419.73	-6.2565	1.4904e-002	-9.3658e-002
6	704.77	-10.564	1.4988e-002	-9.4183e-002

0.1 damping ratio

Mode	Damped Frequency [Hz]	Stability [Hz]	Modal Damping Ratio	Logarithmic Decrement
1	152.02	-5.1625	3.3939e-002	-0.21337
2	355.63	-17.699	4.9706e-002	-0.3127
3	367.77	-18.342	4.9811e-002	-0.31336
4	415.17	-20.629	4.9627e-002	-0.3122
5	420.2	-20.832	4.9515e-002	-0.31149
6	705.59	-35.176	4.9791e-002	-0.31323

0.2 damping ratio

Mode	Damped Frequency [Hz]	Stability [Hz]	Modal Damping Ratio	Logarithmic Decrement
1	152.46	-10.276	6.7251e-002	-0.42351
2	356.94	-35.268	9.8328e-002	-0.62082
3	369.13	-36.549	9.8533e-002	-0.62213
4	416.7	-41.106	9.817e-002	-0.61981
5	421.73	-41.512	9.7959e-002	-0.61847
6	708.23	-70.097	9.8482e-002	-0.62167

0.3 damping ratio

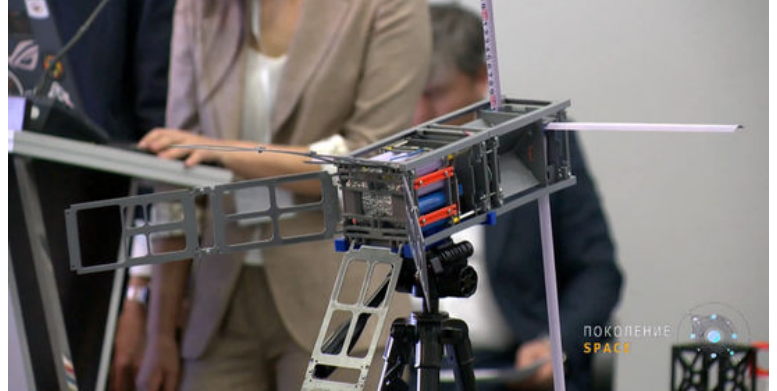
Mode	Damped Frequency [Hz]	Stability [Hz]	Modal Damping Ratio	Logarithmic Decrement
1	153.16	-15.297	9.938e-002	-0.62753
2	359.06	-52.587	0.14491	-0.92021
3	371.34	-54.488	0.14521	-0.92214
4	419.19	-61.231	0.14467	-0.91867
5	424.22	-61.901	0.14439	-0.91682
6	712.5	-104.52	0.14515	-0.92175

ABYLAIKHAN MUKHAMEJANOV

MECHANICAL ENGINEERING AT LEHIGH UNIVERSITY

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SKILLSAT STARTUP



What?

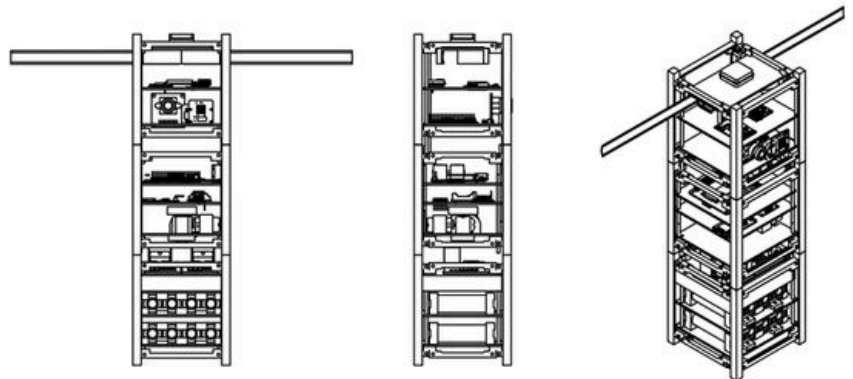
- Founded SkillSat, an EdTech startup to manufacture STEM kits that teach students to build stratospheric satellites.

How?

- Used Arduino and Raspberry Pi as the base for satellite systems.
- Utilized additive manufacturing to prototype 25+ designs.
- Applied DFA and DFM principles to reduce production costs by 44%.
- Tested prototypes at the National Space Center of Kazakhstan.

Results

- Launched satellites to Near Space (40 km+) via stratospheric balloons.
- Raised \$15K in funding and distributed 80+ kits to 240+ students across 25+ schools.
- Recognized as a Top-3 SpaceTech and Top-15 EdTech Startup by Forbes Kazakhstan.



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Uncovering the Fluid-Structure Interactions of Wave-Assisted Propulsion via Cyber-Physical Fluid Dynamics



Abylaikhan Mukhamejanov¹, Ata Tankut Ardic¹, Keith W. Moored¹
¹Department of Mechanical Engineering & Mechanics, Lehigh University, Bethlehem, PA 18015

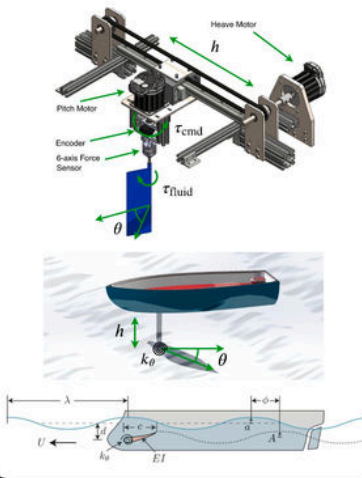
78th Annual Meeting APS DFD
Nov. 23–25, 2025, Houston, TX

Project Objectives and Goals

- Investigate wave-assisted propulsion (WAP) using controlled heave and passive pitch to transform wave energy into forward thrust and improve stability.
- Develop and validate a cyber-physical testbed that couples a flapping hydrofoil with virtual mass-spring-damper dynamics.
- Implement real-time control of virtual stiffness, damping, and inertia to emulate passive pitching behavior.

Motivation

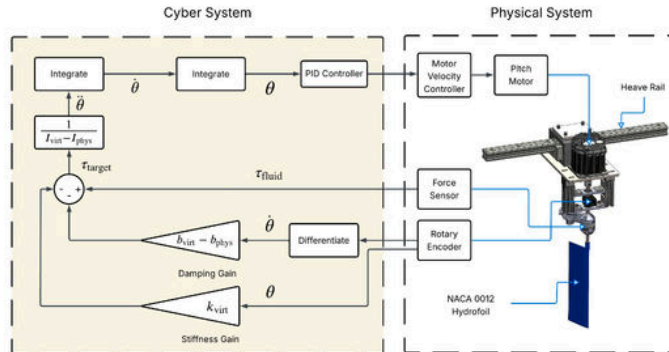
- Wave-assisted propulsion (WAP) uses oscillating hydrofoils to extract energy from waves and generate forward thrust.
- The flapping motion can also reduce vessel oscillations, improving stability and efficiency in marine systems.
- Studying these fluid-structure interactions (FSI) requires precise control of mass, stiffness, and damping parameters.
- Traditional experiments fix mass, stiffness, and damping through mechanical design, limiting flexibility in studying dynamic behavior.



Methods & Procedures

- A cyber-physical system [1] couples the physical actuator-fluid testbed with a computer-based controller that enforces a virtual dynamic model in real time.
- Command signal cancels physical inertia and damping to match desired virtual parameters.
- Physical inertia and damping identified via ring-down experiment.
- Double integration [2] of force/torque removes filtering need and enables stable 4 kHz Simulink Real-Time control loop.

$$\tau = I \ddot{\theta} + b \dot{\theta} + k \theta$$
$$\tau_{\text{virt}} = - \left(I_v \ddot{\theta} + b_v \dot{\theta} + k_v \theta \right)$$
$$I_p \ddot{\theta} = \tau_{\text{fluid}} + \tau_{\text{virt}}$$
$$(I_p + I_v) \ddot{\theta} + (b_p + b_v) \dot{\theta} + k_v \theta = \tau_{\text{fluid}}$$



Experimental Setup

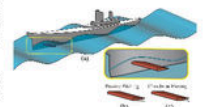


Conclusions

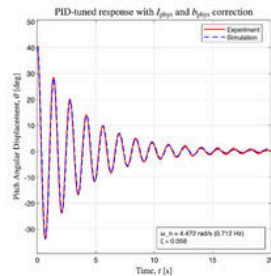
- Developed a working cyber-physical system mimicking passive pitching.
- Any structural parameters can be virtually assigned to the system, including nonlinear stiffness elements and damping characteristics.
- The system accepts real heave motion inputs from naval vessels, enabling experimental investigation and optimization of Wave-Assisted Propulsion (WAP).

Future Studies / Recommendations

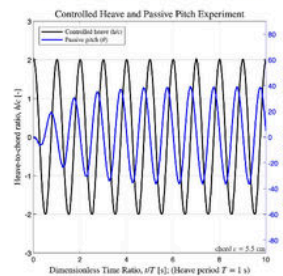
- Evaluate propulsion efficiency and stability under varying environmental and structural parameters in the water channel.
- Recreate natural sea-state conditions by coupling heave-wave excitation with vessel motions across multiple degrees of freedom.



System Validation



Air Experiment Results



Acknowledgments

This research was supported by the Lehigh Opportunity Fund and Rossin Research Scholars Program (2025), managed by Prof. Mark Snyder, Associate Dean for Graduate Education, Lehigh University.

References

- [1] Mackowski, A. W., and Williamson, C. H. K., 2011, "Developing a Cyber-Physical Fluid Dynamics Facility for Fluid-Structure Interaction Studies," *J. Fluids Struct.*, 27 (5–6), pp. 748–757.
- [2] Zhu, Y., Su, Y., and Breat, K., 2020, "Nonlinear Flow-Induced Instability of an Elastically Mounted Pitching Wing," *J. Fluid Mech.*, 899, A35.